

Dynamic Programming - 7

② Subset Sum

Given an array of N elements, check

① If there is a subset whose sum

② is K calc how many subsets do we have?

arr[] = { 1, 5, 7, 6, 2, 0 } n

$K = 8$

6 subsets

1 { 1, 7 } { 1, 7, 0 }
2 { 1, 5, 2 } { 1, 5, 2, 0 }
3 { 6, 2 } { 6, 2, 0 }

★ [22-Dec (Language Module)
Townhall

✓ → Prepone 22nd Dec DSA class

(M/W/F)

✓ → Tree 6 : Deletion in Trees /

↓
Schedule

x → Extra Class ?

subset Sum

$\boxed{1, 5, 7, 6}, 2$

$$f(a, n, k=8)$$


inc 2

+

exc

①

$$\Rightarrow f(a, n-1, 6)$$
$$f(a, n-1, 8)$$

1576

inc 6

$$f(a, n-2, 0)$$

Base Case

$\boxed{157} \times$

$$f(a, n-2, \underline{6})$$

TODO

 inc

0

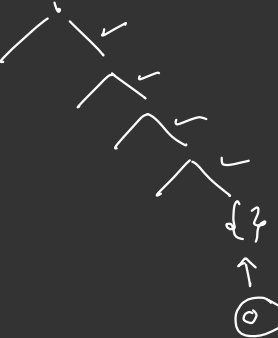
↓
(1, 5)

$$f(\underline{a}, n-3, 6)$$

1

$$f(a, n-4, 1)$$

A hand-drawn diagram of a square with a vertical line through the center, labeled '1'.

$$f(a, n-4, 6)$$
$$f(a, n-5, 0)$$
$$f(a, n-5, 1)$$
$$f(a, a-5, 5)$$
$$f(9, 4-5, 6)$$


Rec way

→ $O(2^N)$

$f(arr, n, k)$ { ^{dp}

if ($k == 0$)
return 1

if ($n == 0$)
return 0

dp[n][k]

// Rec. way

inc = $f(arr, n-1, k - arr[n-1])$ ^{dp}

exc = $f(arr, n-1, k)$ ^{dp}

return $dp[n][k]$
inc + exc

if $arr[n-1] \leq k$
↑

Unique States

↓

$N \rightarrow N+1$ states

$K \rightarrow 0 \dots K \Rightarrow K+1$ states

↓

$O(NK)$ using Top Down / BU DP

Overlapping

Assume the elements and +ve K

$n \times K$

0 elements ↓

⇒ $n=0$

① $n=1$

⑤ $n=2$

⑦ $n=3$

⑥ $n=4$

② $n=5$

	K=0	K=1	K=2	K=3	K=4	K=5	K=6	K=7	K=8
n=0	1	0	0	0	0	0	0	0	0
n=1	1								
n=2	1								
n=3	1								
n=4	1								
n=5	1								

0 1 2 3
1, 5, 7, 6, 2

ans

$$dp[0][0] = 1$$

↓ 3

1 based

$$dp[i][j] =$$

i elements

inc

exc

$$dp[i-1][j - arr[i-1]] + dp[i-1][j]$$

0 based

$dp[4][6]$
 ↓
 upto 4 elements
 ↑
 6 sum

$$j - arr[i-1] \geq 0 \Rightarrow arr[i-1] \leq j$$

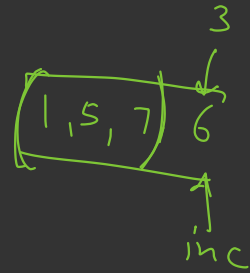
$$i=4 \quad j=6-6=0 \Rightarrow dp[i-1][j-arr[i-1]]$$

$$dp[3][0]$$

↓
①

$n, k \rightarrow$ variables to iterate over dp array

$$dp[n][k] = dp[n-1][k - arr[n-1]] + dp[n-1][k]$$

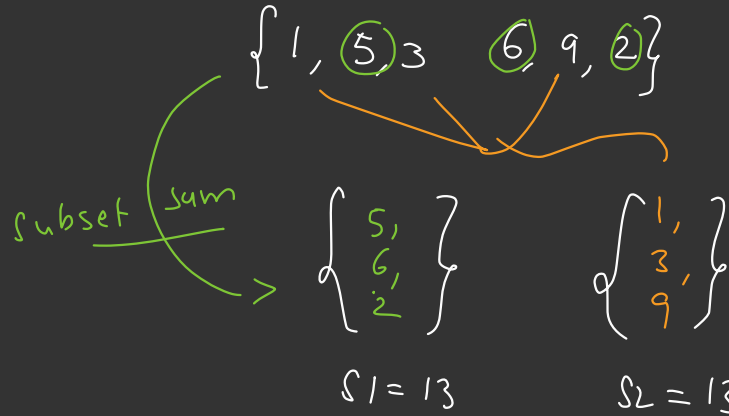


Q2

Given an array containing N elements, divide all elements in 2 subsets such that both subsets have equal sum.

Note: Each element $a[i]$ must belong to exactly one of the subset.

Output **yes/or no** if such a subset is possible.



$$\text{Sum} = 1 + 5 + 3 + 6 + 9 + 2$$

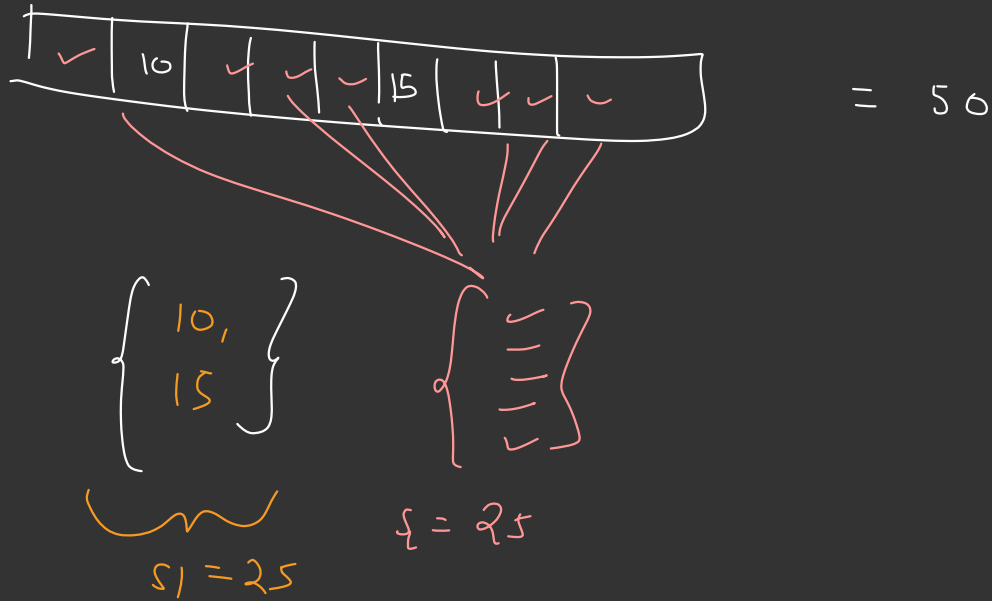
$$= 26$$

↑
even

$$27$$

↑
odd $\Rightarrow$ NO

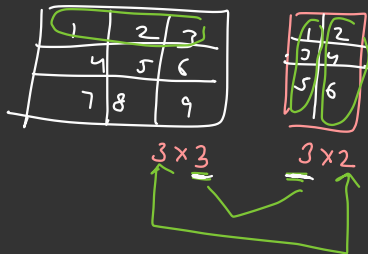
subset with $\text{sum} = \frac{\sum \text{all}}{2}$



- 1) Calc total sum, if sum is Odd Output No.
- 2) If sum is even, then solve subset sum problem with $K = \text{sum}/2$

→ Matrix Chain Multiplication

Concept →



$$\underline{1 \times 1} + \underline{2 \times 3} + 3 \times 5$$

$$\underline{1 \times 2} + \underline{2 \times 7} + \underline{3 \times 6} = 2 + 8 + 18$$

$$= \begin{bmatrix} 22 & 28 \\ \checkmark & \checkmark \\ \checkmark & \checkmark \end{bmatrix}$$

3x2 values
No of mut = $3 \times 3 \times 2$

m x p

p x n

$$= \begin{bmatrix} & \\ & \end{bmatrix}$$

m x n

→ $m \times n \times p$ multiplications (Standard)

list of Matrices $\rightarrow$ Reduce the no of multiplications

$$\left(\begin{array}{cc} M1 & M2 \\ \underline{3 \times 5} & \underline{5 \times 7} \end{array} \right)^x \left(\begin{array}{cc} M3 & M4 \\ \underline{7 \times 4} & \underline{4 \times 3} \end{array} \right)^x$$

$$= \boxed{\quad}^{3 \times 3}$$

$$\left(\begin{array}{cc} \underline{M1} & \underline{M2} \\ \underline{\quad} & \underline{\quad} \end{array} \right) \left(\begin{array}{cc} \underline{M3} & \underline{M4} \\ \underline{\quad} & \underline{\quad} \end{array} \right)$$

$$\downarrow$$
$$3 \times 5 \times 7$$

$$= 105$$

$$7 \times 4 \times 3$$

$$= 84$$

$$\left(\begin{array}{c} 3 \times 7 \\ \hline A \end{array} \right) \left(\begin{array}{c} 7 \times 3 \\ \hline B \end{array} \right) = 3 \times 7 \times 3$$
$$= 63$$

$$\begin{array}{r} 105 \\ + 84 \\ + 63 \\ \hline 252 \end{array}$$

$$\begin{array}{cccc}
 \overline{M_1} & M_2 & \overline{M_3} & M_4 \\
 \underline{3 \times 5} & \underline{5 \times 7} & \underline{7 \times 4} & \underline{4 \times 3}
 \end{array}$$

4 mul

$$\begin{array}{c}
 \text{Input} \\
 \hline
 \xrightarrow{\text{Inputs}}
 \end{array}
 \left[\begin{array}{cccc}
 3, & 5, & 7, & 4, & 3 \\
 3 \times 5 & 5 \times 7 & 7 \times 4 & 4 \times 3 &
 \end{array} \right]$$

$$\begin{array}{cc}
 M_1 & M_2 \\
 \left[\begin{array}{c} M_{12} \\ 3 \times 7 \end{array} \right] & \left[\begin{array}{c} M_3 \\ 7 \times 4 \end{array} \right] \\
 \searrow & \\
 \left[\begin{array}{c} M_{123} \\ 3 \times 4 \end{array} \right] & \left[\begin{array}{c} M_4 \\ 4 \times 3 \end{array} \right]
 \end{array}$$

$$= 3 \times 5 \times 7 = 105$$

$$= 3 \times 7 \times 4 = 84$$

$$= 3 \times 4 \times 3 = 36$$

225 ← lesser

$$a(i) \rightarrow a(i) \times a(i+1)$$

Which way is better

$$\left\{ \begin{array}{l}
 - \left((M_1 \ M_2) (M_3 \ M_4) \right) \\
 - \left(M_1 \left((M_2 \ M_3) \ M_4 \right) \right) \\
 - \left(\left(M_1 \ (M_2 \ M_3) \right) M_4 \right) \\
 \vdots
 \end{array} \right.$$

$$N = 5$$

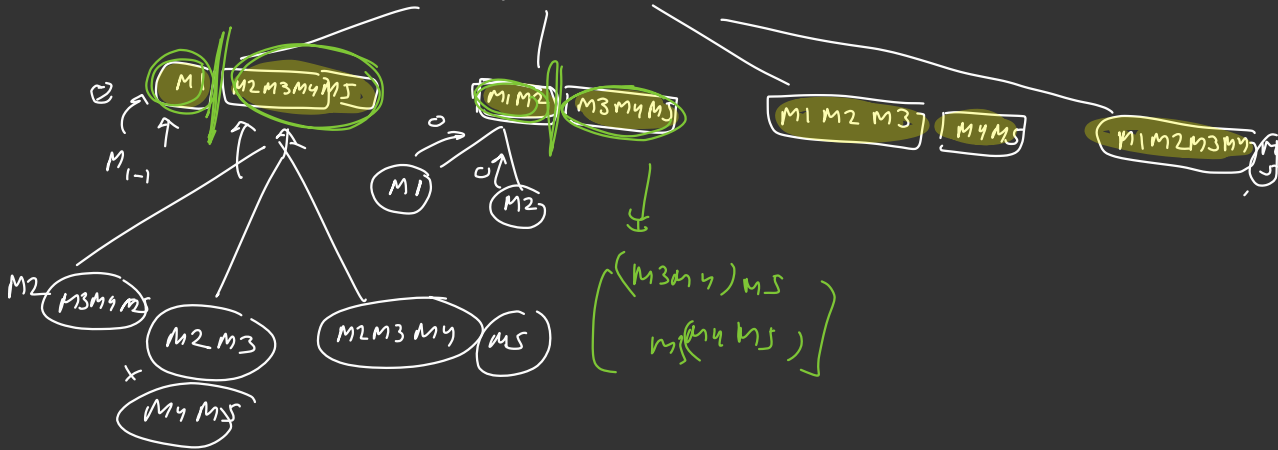
$$\rightarrow [2 \ 3 \ 4 \ 1 \ 7 \ 3]$$

min calc will be needed

multiply ($i=1, j=5$)

$$\left\{ \begin{array}{l} 1^{st} \ M_1 = 2 \times 3 \\ M_2 = 3 \times 4 \\ M_3 = 4 \times 1 \\ M_4 = 1 \times 7 \\ M_5 = 7 \times 3 \end{array} \right.$$

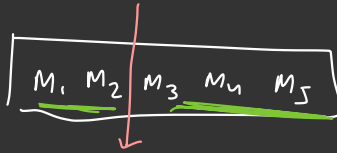
$$out \rightarrow \underline{2 \times 3}$$



$\text{multiply}(i, j) = \{ \text{min cost to multiply all matrices from idx } i \text{ to } j \}$

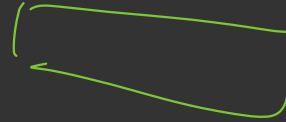
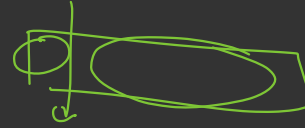
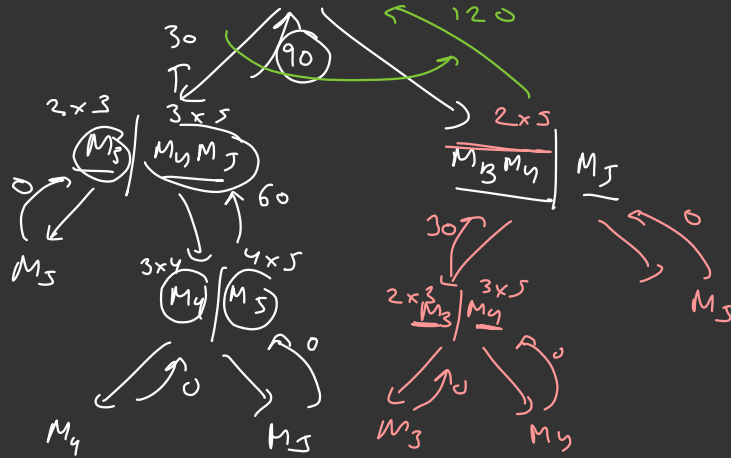
$$\begin{array}{l}
 \begin{array}{c} 3 \times 4 \\ \textcircled{M_i} / M_{i+1} \dots M_j \end{array} \begin{array}{c} 4 \times 5 \\ \text{---} \end{array} = \text{min} \left\{ \begin{array}{l} \text{multiply}(i, i) + \text{multiply}(i+1, j) + C_1 \\ \text{multiply}(i, i+1) + \text{multiply}(i+2, j) + C_2 \\ \text{multiply}(i, i+2) + \text{multiply}(i+3, j) + C_3 \\ \vdots \\ \text{multiply}(i, j-1) + \text{multiply}(j, j) + C \end{array} \right\}
 \end{array}$$

$(M_i \dots M_{i+1}) (M_{i+2} \dots M_j)$
 $(M_i \dots M_{i+2}) (\quad)$
 $(M_i \dots M_{j-1}) (M_j)$



$M_1 M_2$

$M_3 M_4 M_5$



$$\text{multiply}(i, j) = \min \left\{ \text{multiply}(i, k) + \text{multiply}(k+1, j) + \right.$$

↑
indices of
array

5 ↓ 3 ↓ 2 ↓ 6

5 3 2 6

5 3 2 6

Base Case

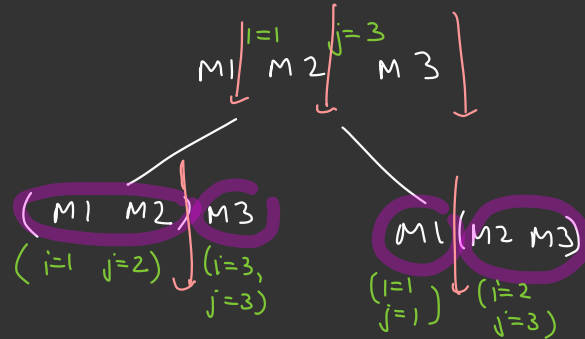
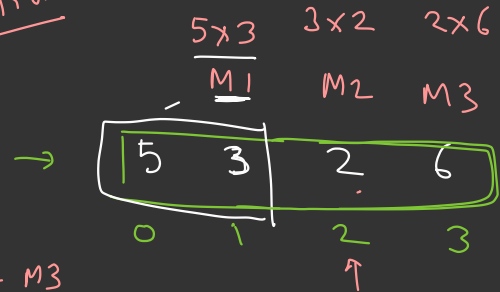
(i=j)

return 0

min cost (1, 3)

↑ ↑
i j

Notation



$\Rightarrow \text{multiply}(\text{arr}, i, j) \{$
Rec Case $\text{min_cost} = \infty$

$\text{for } (k = i; k < j; k++) \{$

$L = \text{multiply}(i, k)$

$R = \text{multiply}(k+1, j)$

$\text{cost} = \underline{a[i-1] \times a[k] \times a[j]}$

$\Rightarrow \text{total_cost} = L + R + \underline{\text{cost}}$

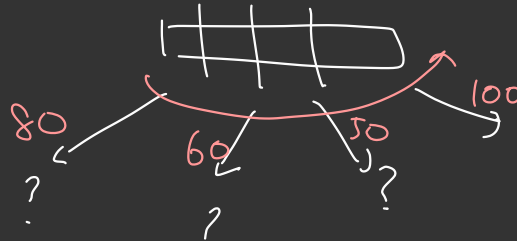
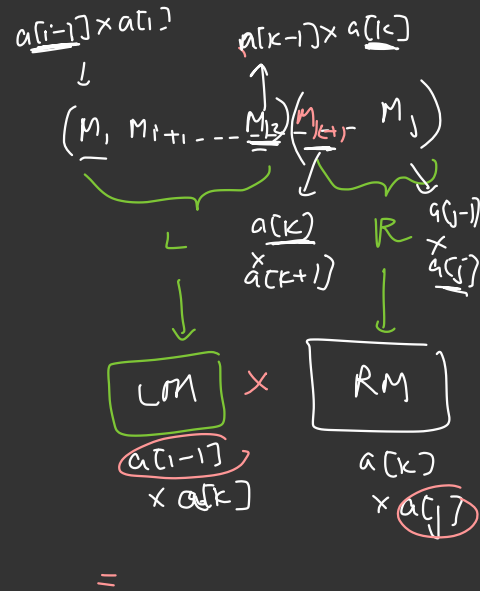
$\text{if } (\text{total_cost} < \text{min_cost})$

$\text{min_cost} = \text{total_cost}$

$\}$

return min_cost

$\}$



multiply (1, N) $\rightarrow$ ans

